SOFTWARE ENGINEERING

LAB ASSIGNMENT 04

Activity Diagram

Digital Wallet Management System (DWMS)

SUBMITTED BY — TEAM 5

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1. Introduction to Activity Diagrams

A UML Activity Diagram models the control flow of a process from its initial node to its final outcome. Rounded action nodes represent activities, arrows show the order of execution, and decision diamonds show conditions that select alternative paths. Activity diagrams are useful for reviewing workflows and ensuring that success, validation failure, cancellation, and external-service failure are considered.

1.1 Objectives

- Model the Add Money workflow specified in the DWMS SRS.
- Represent validation and user-confirmation decisions explicitly.
- Show the successful funding path and the unsuccessful path without updating the wallet incorrectly.

2. Selected Scenario: Add Money to Wallet

The user opens the wallet dashboard, chooses Add Money, enters an amount and funding method, and reviews a summary. After explicit confirmation, the DWMS sends a request to a bank or payment gateway. The wallet balance is updated only when the external funding result is confirmed successful.

2.1 Relevant SRS Requirements

- REQ-WAL-3: allow the user to add money using a supported funding method.
- REQ-WAL-4: validate the funding request and require explicit confirmation.
- REQ-WAL-5: update the wallet balance only after successful confirmation from the bank/payment gateway.
- REQ-NOTIF-1: notify the user when a transaction affecting the wallet completes, whether successfully or unsuccessfully.

UML DIAGRAM

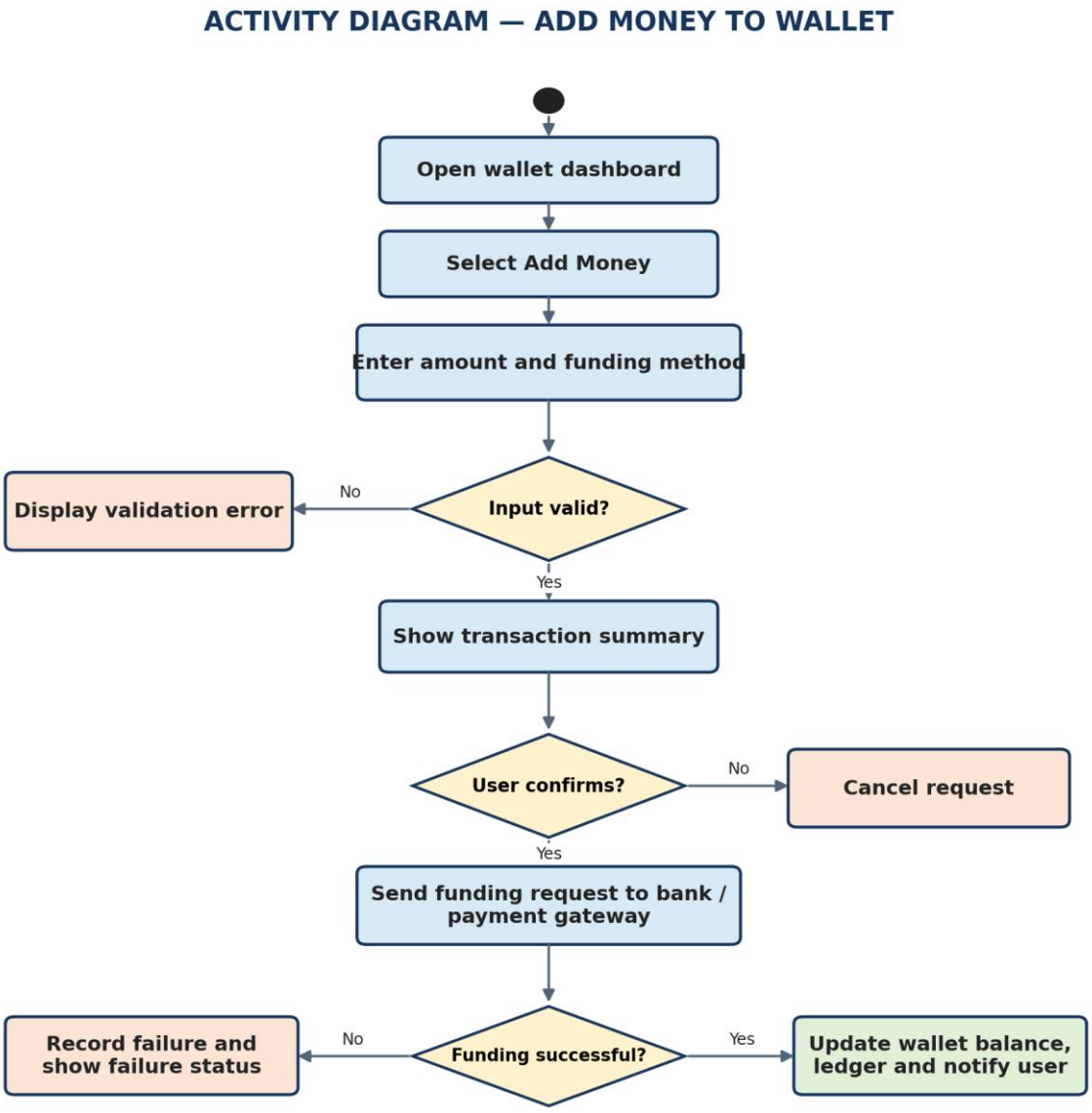


Figure 1. Activity diagram for adding money to a digital wallet.

3. Workflow Explanation

1. The workflow starts when the authenticated user opens the wallet dashboard and selects Add Money.
2. The user enters the amount and selects a supported funding method. The system validates the input.
3. If the input is invalid, an error is displayed and the request does not proceed to funding.
4. If valid, the system displays a transaction summary and waits for explicit confirmation. The user may cancel without committing a financial operation.
5. After confirmation, the DWMS sends the funding request to the external bank/payment gateway.
6. If funding succeeds, the system updates the wallet balance and ledger, then notifies the user. If funding fails, it records/reports failure and does not credit the wallet.

4. Design Considerations

- Explicit confirmation prevents accidental financial actions.
- External-service responses determine whether the wallet balance can be updated.
- Failure paths must leave the ledger and wallet consistent.
- Notification channels are configurable and remain a TBD item in the SRS.

Conclusion

The activity diagram captures both the normal Add Money path and its important alternative outcomes. It makes validation, confirmation, external processing, and balance-update conditions visible before implementation and supports deriving test cases for invalid input, cancellation, gateway failure, and successful funding.